

External HMI for self-driving vehicles: Which information shall be displayed?



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ABSTRACT

The objective of this study was to investigate pedestrians' informational needs towards self-driving vehicles (SDVs). Previous research has shown that external human-machine interfaces (eHMIs) compensate for pedestrian-driver communication when SDVs are integrated into traffic. However, detailed insights on which information the eHMI shall provide lack so far. In a mixed design study, $N = 59$ participants encountered a simulated driverless vehicle in different traffic scenarios (a. unsignalized intersection vs. b. parking lot; between-subject factor). We investigated the effect of no eHMI (baseline) vs. eHMIs displaying the automated driving system (ADS) status, and informing subsequently about its perception of the pedestrian and/or its intent for the next maneuver ((1) no eHMI, (2) status eHMI, (3) status + perception eHMI, (4) status + intent eHMI, (5) status + perception + intent eHMI; within-subject factor). A mixed-methods design was used to explore participants' subjective feelings, traffic behavior, and underlying attitudes. The findings reveal that any eHMI contributes to a more positive feeling towards SDVs compared to the baseline condition without eHMI, consistent among traffic scenarios: participants felt significantly safer, reflected greater trust and user experience ratings, and perceived the SDV as more intelligent and transparent. The status indicator mainly drives these beneficial effects on subjective measures. Participants reported that the status information explains the absence of a driver steering the vehicle. Compared to the status eHMI, the status + perception eHMI reflects no further benefit regarding subjective feelings and even has a negative impact on traffic flow. Moreover, participants regarded the additional information on the vehicle's perception as an obvious gimmick. On the contrary, the status + intent eHMI increases user experience, perceived intelligence, and transparency for pedestrians more than the mere status eHMI. Participants reported that additionally informing about the vehicle's intent adds a further sense of safety. The present study failed to show any improvements in traffic flow but found evidence for individual crossing and clearing strategies among pedestrians. This work can inform the future design of eHMIs.

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1. Introduction

In today's traffic, pedestrians rely on different communication strategies when encountering a vehicle, including vehicle speed (Ackermann, Beggiato, Bluhm, Löw, & Krems, 2019b; Petzoldt, Schleinitz, & Banse, 2018; Šucha, Dostal, & Risser, 2017), distance of the vehicle (Liu & Tung, 2014), and eye contact with the driver (Šucha et al., 2017). While interpreting a vehicle's kinematics can be seen as a key communication strategy (Dey & Terken, 2017; Šucha et al., 2017; Mahadevan, Somanath, & Sharlin, 2018), additional informal communication like eye contact, hand gestures, and body movements become especially prominent in ambiguous situations that are not strictly defined by legal regulations, such as at a parking lot (Färber, 2016; Witzlack, Beggiato, & Krems, 2016). For instance, a sweeping hand gesture or a palm facing upwards signify a gesture of offering and giving right of way (Färber, 2016). With vehicles of SAE Level 4 (high automation) and 5 (full automation), a driver is no longer required (SAE, 2018). Previous studies showed that pedestrians are uneasy when losing the possibility to communicate with a driver (Lagström & Lundgren, 2015; Lundgren et al., 2017; Rodríguez, 2017). Thus, it is critical to examine how SDVs can smoothly interact with pedestrians, substituting communication with the driver, particularly in complex urban environments where informal communication strategies are used frequently today (Visser, van der Kint, van Schagen, & Hagenzieker, 2016). To address this issue, the application of an external human-machine interface (eHMI) for SDVs has been proposed. An eHMI shall provide information to other traffic participants, improving the efficiency of the encounter between an SDV and a pedestrian (Schieben et al., 2019).

While various studies show a beneficial effect of eHMIs on pedestrian-vehicle interaction, the question which information an eHMI should display is not fully answered yet. We conducted a Wizard-of-Oz study to compare the effectiveness of the following eHMI information displays: no eHMI (baseline), engagement of the automated driving system (ADS), the vehicle's perception of the pedestrian and the vehicle's intent for the next maneuver.

In the following, an overview of preceding eHMI studies is given. Subsequently, we present the research gap addressed by the current study and eHMI design considerations. Finally, research questions are listed.

1.1. Preceding eHMI studies

So far, various eHMI concepts have been created for SDV-pedestrian communication. These eHMI concepts can be divided into four categories of conceivable messages displayed (Schieben et al., 2019): (1) Informing about the vehicle's ADS status: pedestrians are able to distinguish between a vehicle steered manually and a vehicle in automated driving mode. (2) Informing about an SDV's perception: pedestrians see that they have been detected by the ADS. (3) Informing about an SDV's intent: pedestrians understand the ADS's intended next movements and can adapt their behavior. For example, the vehicle could indicate that it is yielding. (4) Giving advice to pedestrians: pedestrians are told explicitly whether they should walk or stop. In the following, work related to each of the conceivable information types is presented.

1.1.1. ADS status

Insights into an eHMI that only displays the engagement of the ADS, but no more information, are limited. In a Wizard-of-Oz field observation, Rothenbücher, Li, Sirkin, Mok, and Ju (2016) observed pedestrians crossing a sidewalk where they have the right of way while a simulated SDV was approaching. The vehicle had "Stanford Autonomous Vehicle" vinyl stickers on the doors and the hood. We argue that this text message can be seen as a form of automated driving mode indication. Pedestrians managed interactions smoothly. Ackermann, Beggiato, Schubert, and Krems (2019a) evaluated different eHMI variants conveying either the vehicle's status or advice to the pedestrian to cross the street. Status information indicated that the vehicle drives in automated mode, for example, as a text message on an LED display. The study participants preferred the concrete advice to status information. In both studies, there was no test condition without eHMI serving as a baseline.

In a Wizard-of-Oz study, Hensch, Neumann, Beggiato, Halama, and Krems (2020) contrasted a baseline test condition without eHMI with various eHMI designs, including a status eHMI (steady light) and an advice eHMI (sweeping light). In a parking area, they interviewed uninformed random pedestrians passing by. Pedestrians did not report any differences between no eHMI and the eHMI signals regarding comprehensibility, trust, subjective safety, and perceived usefulness. The authors conclude that light-based eHMI signals are not comprehensible by intuition and, thus, road users need to be informed about signal meaning beforehand (as already suggested by Habibovic et al., 2018).

1.1.2. Perception

In a study by Merat, Louw, Madigan, Wilbrink, and Schieben (2018), people were surveyed that had interacted with an automated transport vehicle of the CityMobil2 project in different European cities. The vehicle was not equipped with an eHMI, but interviewees were asked what kind of information the vehicle should convey to ensure that they feel safe around it. Overall, participants evaluated an acknowledgement that they have been detected as most important, followed by communicating intent. Motivated by the fact that eHMIs shall replace current pedestrian-driver interaction relying on eye contact, Chang, Toda, Sakamoto, and Igarashi (2017) proposed a concept for anthropomorphizing the SDV by attaching moving stylized eyes as an interface in the frontal headlights. These convey the SDV's perception of the pedestrian by looking at him/her if the vehicle is going to stop. The concept was tested in a virtual reality study in a zebra crossing situation. Compared to no eHMI, pedestrians felt safer and made their street crossing decision faster in the presence of the eyes.

1.1.3. Intent

Mahadevan, Somanath, and Sharlin (2018) compared different eHMI prototypes including LEDs and a display to communicate the vehicle's perception and intent. For instance, an animated face on a display with moving eyes conveyed the vehicle's awareness of the pedestrian, similar to the idea proposed by Chang et al. (2017). Moreover, moving purple lights indicated that the vehicle is about to start. Mahadevan et al. (2018) findings reveal that participants find cues communicating intent more important than those displaying perception.

In a virtual reality study, Böckle, Brenden, Klingegård, Habibovic, and Bout (2017) proposed an eHMI communicating its intents to start driving, to (not) stop, and to wait by using different colors and light motions. For example, yellow flashing LEDs conveyed the vehicle's intent for not stopping. After having learned the meaning of the signals, participants felt safer and more comfortable crossing the street when the eHMI gave information on the vehicle's intent compared to encounters without the eHMI engaged. Furthermore, some participants reported that the eHMI helped them to cross the street earlier, whereas some participants were more hesitant to cross and waited until the vehicle came to a complete standstill without the eHMI. Habibovic et al. (2019) have proposed a similar eHMI concept. An LED strip attached to the front windshield displays the engagement of the ADS by a steady light and its intent to yield as well as to start driving by light motion. These displays are activated sequentially in a traffic situation. This eHMI concept was evaluated in a series of field studies, and the findings suggest that communicating the vehicle's intent to pedestrians contributes to a positive experience, perceived safety, and calmness when encountering an SDV. In a virtual reality study, de Clercq, Dietrich, Núñez Velasco, de Winter, and Happee (2019) contrasted a baseline test condition without eHMI with various eHMI designs, including a text-based advisory eHMI "Walk"/"Don't Walk" and a light-based eHMI displaying the vehicle's intent to yield via frontal brake lights. The usage of different modalities (text vs. light) to compare different information (advice vs. intent) constitutes a limitation of their study. Participants were standing on the pavement next to a two-lane road, watching approaching vehicles. Results showed that any eHMI improves pedestrians perceived safety to cross, compared to no eHMI. There were no significant differences in feeling safe to cross between the eHMI test conditions. However, the authors point out that text is unambiguous and does not have to be learned, contrary to the other eHMI designs.

1.1.4. Advice

Most previous eHMI studies were conducted with an advice-based interface. Song, Lehsing, Fuest, and Bengler (2018) examined two advice-based eHMIs in comparison to a baseline condition without an eHMI. The interfaces showed either "Go!" or "OK!" as text message complemented by a graphic element, the German zebra crossing sign, to indicate to pedestrians that it was safe to cross the road. Participants saw videos of an approaching vehicle and had to decide whether to cross the road (pressing space key) or not to cross (not pressing space key). The results reveal that participants indicated to cross the road more often when any advisory eHMI was present compared to when no eHMI was present (63% vs. 22%). There was no difference between the two eHMI designs. A study by Hudson, Deb, Carruth, McGinley, and Frey (2019) using a virtual reality setup at a crosswalk contrasted a no-eHMI test condition and various advice-based eHMI designs. These eHMI designs included the visual text display "walk," an auditory text message "safe to cross," a walking silhouette, and a stop sign for unsafe crossing. Results show that any eHMI is rated better than no eHMI and that text, whether visual or auditory, is the preferred feature to indicate safe crossing conditions. A virtual reality study by Stadler, Cornet, Theoto, & Frenkler (2019), moreover, shows that eHMI symbols advising the pedestrian to cross or not to cross increased the efficiency and effectiveness of the interaction with an SDV in an ambiguous crossing situation compared to encounters without eHMI. For example, they used a green walking man to advise pedestrians to cross and a red standing man to advise pedestrians not to cross, adapted from traffic lights. Their results show that an eHMI helped pedestrians to make their street crossing decision faster and with fewer errors; that is, pedestrians only crossed when the vehicle really stopped for them.

1.2. Research gap and design considerations

Previous studies show that different information displays lead to a positive effect, i.e. conveying a vehicle's perception (Chang et al., 2017), a vehicle's intent (Böckle et al., 2017; Habibovic et al., 2019), and a vehicle's advice (Hudson et al., 2019; Song et al., 2018; Stadler et al., 2019). However, only few studies reached conclusions on the concrete information an eHMI shall display. Based on their findings, Mahadevan et al. (2018) emphasize that information on intent is regarded as being more important than perception. In contrast to that, the majority of participants in Merat et al. (2018) study rated perception information as being most important, followed by intent. In Ackermann et al. (2019a), results point to a preference for advice over status information.

Given that previous research has not achieved unanimous and comprehensive results regarding the information an eHMI shall display, we addressed two research gaps with the present study: (1) To the best of our knowledge, no study has conducted a comprehensive comparison of different eHMI concepts with a focus on the conceivable messages. From previous work, it can be concluded that pedestrians want to receive information from SDVs, but not *which* information. (2) Secondly, to the best of our knowledge, there exists no systematic comparison between no eHMI and an eHMI displaying the ADS status. Those studies that incorporated an "ADS status indicator" test condition lacked a baseline test condition without eHMI (Ackermann et al., 2019a; Rothenbücher et al., 2016) or failed to inform participants about the meaning of the eHMI signal (Hensch et al., 2020). Thus, the positive effects of any eHMI displaying a vehicle's intent, perception, or advice shown in

previous studies might be confounded with the inherently provided information that the vehicle is operating in automated driving mode.

However, for the current study we deliberately decided to not include an advice-based eHMI test condition. An eHMI that explicitly provides advice, instructions, or guidance to a pedestrian, for example, to cross the street, is problematic from a legal and ethical point of view. As pointed out by the ISO (2018) and Andersson, Habibovic, Klingegård, Englund, and Lundgren (2017), SDVs cannot necessarily account for surrounding traffic that could pose a danger to the pedestrian, for instance, a vehicle travelling in the opposite direction. Advice would imply that it is actually safe to cross the street. This is especially problematic when eHMIs use traffic signs like traffic lights (Stadler et al., 2019), a zebra crossing sign (Song et al., 2018), or a stop sign (Hudson et al., 2019). Hence, we argue that SDVs should not recommend actions for other traffic participants, but only indicate their own ADS status, perception, and/or intent.

Though previous studies showed that text messages are the most preferable interface (e.g. Ackermann et al., 2019a; Hudson et al., 2019), we chose to display all information with light signals. Text requires language skills and is therefore not internationally applicable. With the goal of a harmonized eHMI among vehicle brands as well as across borders, standardization organizations propose light signals for eHMI design (ISO, 2018; SAE, 2019; UNECE, 2019). Conventional vehicles already make use of light signals to inform their surrounding traffic, for example, with brake lights or turn indicators. SAE (2019) proposes to use blue-green light as it is a highly visible and novel color in traffic and is not yet associated with a specific meaning (Dietrich, Willrodt, Wagner, & Bengler, 2018; Werner, 2018). The limitation to blue-green light led to our decision to use different light signals to convey different information. For the ADS status information, we chose a steady light as recommended by the SAE (2019). The human visual system is more sensitive to moving and flashing lights because it is primed to respond to change (Fuller, 2005). Thus, these attention-attracting signals can be used to draw attention to a system change (Endsley, 2012) such as the SDV's perception of a pedestrian and/or the SDV's intent to yield.

1.3. Research questions

The aim of this work is to complement and extend previous studies by investigating how a light-based eHMI affects pedestrian-SDV interaction. With this in mind, we focused closely on the question of *which* information a pedestrian would like to receive from SDVs; that is, feedback pertaining to the vehicle's status of automated driving, plus possibly the ADS's perception of the pedestrian near the vehicle and/or its intent for the next maneuver. Thus, the present study explored five eHMI variants: (1) no eHMI, (2) status, (3) status + perception, (4) status + intent, and (5) status + perception + intent. The status eHMI informing about the engagement of the ADS can be seen as the most basic information, thus it was the basis of all eHMI designs. We conducted a study on a testing site that compares and evaluates the five eHMI variants (within-subject factor) with regard to how participants acting as pedestrians experience interaction with a simulated automated vehicle. To increase generalizability of the results, we chose two urban scenarios (between-subject factor), namely an unsignalized intersection, where a pedestrian wants to cross the street while the vehicle is approaching, and a parking lot, where the vehicle wants to leave a parking space and is blocked by a pedestrian, who has to step aside.

The current study used a mixed-methods approach. This includes participants' subjective evaluations after each trial regarding trust, perceived safety, user experience, perceived intelligence, and transparency. Retrospective interviews explore reasons for and against receiving each item of information. Furthermore, on-site camera recordings are used to analyze the reaction time of pedestrians, to investigate whether an eHMI contributes to an optimized traffic flow as suggested by previous virtual reality studies (Böckle et al., 2017; Chang et al., 2017; de Clercq et al., 2019; Stadler et al., 2019).

2. Method

2.1. Participants

Fifty-nine participants (32 male, 27 female), ranging in age from 18 to 66 years ($M_{age} = 42.98$ years, $SD_{age} = 14.85$ years), constituted the final sample. Data from $N = 7$ participants was excluded due to insufficient German language skills (3), lack of belief in the vehicle's ADS (3), and unwillingness to fulfil the task as instructed (1). Participants had no constraints in personal mobility, normal or corrected-to-normal vision, and no self-reported color blindness. Participants were randomly assigned to the two test groups "intersection" ($N = 30$) and "parking lot" ($N = 29$), controlling for age, $t(57) = -0.01$, $p = .993$, and gender, $\chi^2(1) = 0.02$, $p = .887$. All participants gave written informed consent and received monetary compensation. The study was approved by the RD Ethical Clearing Committee of Daimler AG.

2.2. Location and apparatus

The field study was conducted at Daimler AG's testing and technology center in Immendingen, Germany. To create a Wizard-of-Oz setup of an ADS, fake sensors were mounted on top of a Mercedes-Benz E-Class (Series W213) (Fig. 1). LED strips were attached to the fake sensors and above the windshield. The driver was wearing a seat costume to give the impression that the vehicle did not have a driver. This method has been shown to reliably create the deception of a driverless



Fig. 1. Wizard-of-Oz vehicle with fake sensors support the deception of an ADS.

vehicle in a proof-of-concept study (Rothenbücher et al., 2016). A staff member sitting in the backseat of the vehicle mimicked a passenger.

In a training session, the professional test track drivers steering the vehicle were instructed to deploy a consistent and defensive driving style, as it is known from SDVs (Hawkins, 2018). The drivers were able to reproduce a standardized drive across participants and experimental conditions. Moreover, the vehicle was driving a maximum of 30 km/h, in accordance with the speed limit of most residential areas in Germany (European Commission, 2018; Federal Ministry of Transport and Digital Infrastructure, 2017). The following provisions were made to minimize the risks for participants in the study: (1) Two professional test track drivers were steering the vehicle. (2) The drivers had adequate working hours with sufficient breaks (two shifts of two hours each day). (3) The drivers were instructed to always yield for the participants and to maintain the usual safety margins as in normal traffic. (4) A staff member sitting in the backseat, not the driver, activated the eHMI variants via an interface. (5) The testing locations were shielded from other traffic operating on the testing site.

2.3. Independent variable: eHMI information

We examined five eHMI variants displaying the engagement of the ADS, the vehicle's perception, and/or the vehicle's intent (Fig. 2). All light signals were presented in blue-green (CIE chromaticity values: $x = 0.15$, $y = 0.38$) and were clearly visible in all weather conditions.

- (1) No eHMI – The no-eHMI test condition served as a baseline. Pedestrians did not receive any information via the interface. The vehicle did not display whether the ADS was engaged.
- (2) Status eHMI – The automated driving mode was indicated by a steady light of 6 LEDs at each sensor. We chose this design since it is in line with SAE's recently published recommended practice J3134 (SAE, 2019), recommending the use of blue-green “marker lamps” to display the engagement of the ADS. Moreover, a steady light to indicate ADS engagement was suggested in the study by Habibovic et al. (2019) as well as by Hensch et al., (2020). The status information represents the most basic information display.
- (3) Status + perception eHMI – Besides displaying the vehicle's status, the 6 LEDs on the fake sensors additionally simulated “eye contact” with the pedestrian. Once a pedestrian planning to cross is detected, both LED elements flashed, moved along the LED strip until they focused on the pedestrian's position, and then followed him/her until he/she crossed the street. This signal indicated the vehicle's perception of the pedestrian, meaning that pedestrians receive information that they have been detected (as suggested by Merat et al., 2018), similar to the eye contact in current human-human interaction. The design of moving stylized eyes was shown to be effective in previous research studies (Chang et al., 2017) and proposed in concept cars of different manufacturers (Audi AG, 2018; Jaguar Land Rover, 2018).
- (4) Status + intent eHMI – In addition to the status indication on the fake sensors, the vehicle indicated its intended movements via 30 LEDs centered on the LED light strip mounted above the windshield. By means of a flashing pattern, the intent signal communicated either yielding (slow flashing with a frequency of 0.5 Hz) or planning to go (fast flashing with a frequency of 4 Hz), following a sinusoidal waveform. The design of flashing lights to convey the SDV's intent was shown to be effective in previous research studies (Böckle et al., 2017; Cefkin, Zhang, Stayton, and Vinkhuyzen, 2019). We chose the frequency of 0.5 Hz to achieve a soothing signal and the frequency of 4 Hz to achieve a prompting signal (Chan & Ng, 2009), yet below the threshold of 5 Hz, which may trigger seizures due to photosensitive epilepsy (Walker & Sander, 2012). For the yielding scenario, the signal was connected to the brake, hence it can be seen as a frontal brake light, which was shown to enable pedestrians to detect a braking maneuver more easily and to enhance road safety (Petzoldt et al., 2018). Similarly, Benderius, Berger, and Lundgren (2018) added a fast flashing light before the vehicle starts to increase safety for their light-based eHMI concept, which was best rated during the 2016 Grand Cooperative Driving Challenge.

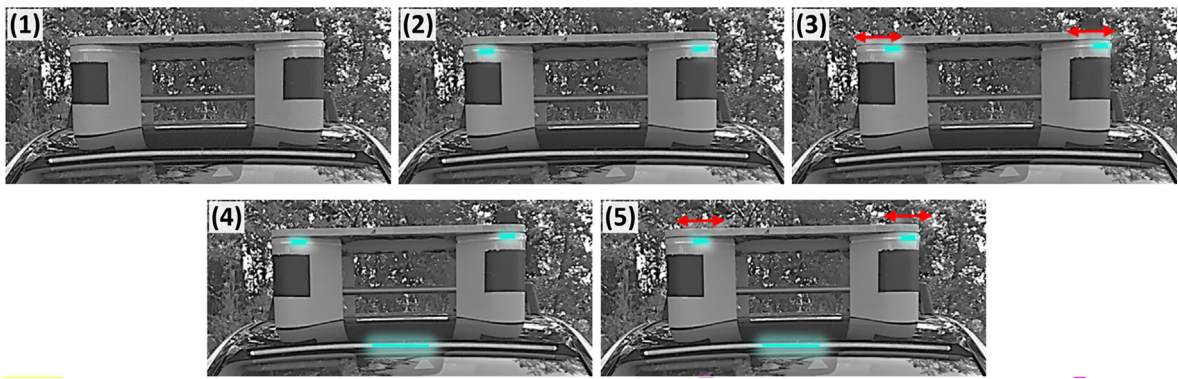


Fig. 2. eHMI test conditions displaying different information to pedestrians. (1) No eHMI: no information is displayed. (2) Status eHMI: steady lights indicate the engagement of the ADS. (3) Status + perception eHMI: additionally to (2), lights on sensors focus on the pedestrian. (4) Status + intent eHMI: additionally to (2), flashing light on windshield indicates vehicle's yielding or planning to go. (5) Status + perception + intent eHMI: combination of (2), (3), and (4).

(5) Status + perception + intent eHMI – This test condition was a combination of all the previous concepts: the engagement of the ADS was indicated by the steady status lights, which then communicated the perception of the pedestrian's position. Furthermore, the vehicle's intent was conveyed via the flashing light above the windshield.

2.4. Procedure

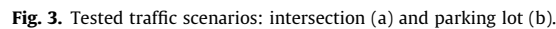
Upon arrival for the study, participants were introduced to the definition of fully automated driving (SAE Level 5). Participants were told that the vehicle they will encounter has a manual and an automated driving mode to clarify that a driver could drive it or it could be self-driving. The study examiner presented the different light concepts and their meanings to the participants by the use of pictures. This procedure assumes that the meanings of the eHMIs are not self-explanatory but have to be learned as a convention, analogous to other signals in traffic, such as traffic lights or traffic signs. All participants encountered each eHMI variant once (within-subject factor) in their corresponding traffic scenario (between-subject factor). The order of the trials was randomized with a Latin square design. To increase generalizability of the results, two test groups in different traffic scenarios were realized: participants encountered the vehicle at either an intersection or a parking lot.

A schematic overview of the intersection traffic scenario is shown in Fig. 3a. Participants were positioned at an uncontrolled and unsignalized four-way intersection. Pedestrians watched a vehicle approaching from a distance of 51 m with a constant speed of 30 km/h (A). At a distance of 25 m from the pedestrian, the vehicle started to yield, indicating that it was planning to turn right (B), to come to a standstill 7 m in front of the pedestrian (C). Thus, the vehicle had an average deceleration rate of $a = -1.93 \text{ m/s}^2$. The participants' task was to cross the intersection as soon as they felt safe to cross. After the pedestrians had crossed the street, the vehicle continued its drive (D). The vehicle yielded for the pedestrians in every trial, as required by traffic laws. If indicated, the status eHMI was activated all the time, as soon as the vehicle came into view (A). If indicated, the perception and intent eHMI were activated when the vehicle started to brake (B) and turned off when the vehicle continued its drive (D).

Fig. 3b gives a schematic overview of the parking lot traffic scenario. The vehicle stood in a parking space between other vehicles. Participants were located at a distance of 3.5 m in front of the parked vehicle (A) and were asked to imagine that the vehicle would be requested by a car-sharing service. At some point, the engine was started (B). The participants' task was to step aside as soon as they felt that the car was planning to set off. After the participant had stepped aside (C), the vehicle rolled out of the parking space and continued its drive (D). If indicated, the eHMI designs were activated as soon as the engine was started (B). If indicated, the perception and intent eHMI were turned off when the vehicle continued its drive (D), while the status eHMI remained engaged.

We chose these scenarios since they are common ambiguous traffic situations in urban areas that require vehicle-pedestrian interaction (Färber, 2016; Witzlack, Beggiato, & Krems, 2016). In Germany, a turning vehicle always has to pay special attention to pedestrians passing in front of it and, if necessary, is obliged to yield for them (Bundesamt für Justiz, 2013). Nevertheless, it is common that drivers do not give right of way to pedestrians. Thus, pedestrians behave risk averse at uncontrolled intersections (Li, Dikmen, Hussein, Wang, & Burns, 2018). For the parking lot situation, no formal rules exist regarding how traffic participants have to behave. However, if the pedestrian does not change his/her position, this leads to a standstill situation since the vehicle detects an obstacle and cannot change its trajectory.

After the study, we applied a two-step manipulation check. First, we asked participants: "Did you notice anything unusual?" and subsequently: "Do you think that the vehicle was driving automated?" Afterwards, participants were fully informed about the setting of the study, and specifically, that the vehicle they encountered had not been driving in automated mode at any time.



2.5.1. Subjective measures

2.5.1. Subjective measures

2.5.2. Observational measures

Video footage of the vehicle encounters was analyzed using the software Mangold Interact. To examine whether an eHMI might contribute to an improvement in traffic flow, two behavioral measures were computed for each traffic scenario: *crossing onset/clearing onset* represents the time in seconds that pedestrians started to cross before the vehicle came to a complete standstill or the time in seconds that pedestrians started to clear the vehicle's trajectory after the engine (and the eHMI) started. We argue that the sooner a pedestrian initiates crossing/clearing, the faster traffic flows. Due to technical difficulties, we could not compute the clearing onset variable for the no-eHMI test condition. For the clearing onset computation, the engagement of the engine and, possibly, the eHMI served as decisive clue(s) for the pedestrians to react. Unfortunately, the engine sound cannot be detected in the video footage, making it impossible to calculate a valid behavioral measurement for the no-eHMI test condition. *Crossing duration/clearing duration* is the time in seconds that pedestrians take to cross the

street (time between pedestrian entering the street and stepping onto the crosswalk on the other side of the street) or the time in seconds that pedestrians take to clear the vehicle's trajectory (time between pedestrian making his/her first step and leaving the vehicle's trajectory). We argue that the faster a pedestrian crosses/clears, the faster traffic flows.

2.5.3. Structured interview

After all trials, a short structured interview followed to derive deeper insights into the experienced eHMI variants. The experimenter posed the following questions successively: "As a pedestrian, do you want SDVs to display... the automated driving mode / that it sees you / that it is yielding or planning to go?" and asked after each question: "Why (not)?"

2.6. Data analysis

We used 2×5 mixed ANOVAs to evaluate pedestrians' subjective ratings of the five eHMI variants (within-subject factor: no eHMI, status eHMI, status + perception eHMI, status + intent eHMI, status + perception + intent eHMI) among two traffic scenarios (between-subject factor: intersection, parking lot). The data was checked for sphericity using Mauchly's test, and where violated, Greenhouse-Geisser and Huynh-Feldt corrections were applied, as recommended by Field (2013). Levene's test for equality of variances was not found to be violated for any measure. For the mixed ANOVAs, we calculated generalized eta squared (η_G^2) as standardized effect size (see Lakens, 2013). As suggested by Bakeman (2005), guidelines by Cohen (1988) can be used for interpretation, where $\eta_G^2 = 0.02$ (small effect), $\eta_G^2 = 0.13$ (medium effect), and $\eta_G^2 = 0.26$ (large effect). For post-hoc *t*-tests, we calculated Pearson's *r* as a standardized effect size measure (as recommended by Field, 2013), which can be interpreted as $r = 0.1$ (small effect), $r = 0.3$ (medium effect), and $r = 0.5$ (large effect) as suggested by Cohen (1992).

For the statistical analysis of the objective measures, we chose to use hierarchical linear modeling (HLM) because it takes into account that variability can be caused either by differences *between persons* due to individual behavioral strategies (as suggested by Clamann, Aubert, & Cummings, 2017), or by variance *within persons* due to the eHMI variants. Extreme cases of the behavioral reactions, defined as more than three times the interquartile range (IQR) greater than the upper or lower quartile, were excluded from data analysis (crossing onset: no exclusions; crossing duration: 1 value excluded; clearing onset: 1 value excluded; clearing duration: 9 values of $N = 6$ participants excluded). Since the eHMI test conditions cannot be treated as a linear trend, each category needs to be calculated against a baseline (Field, 2013). The no-eHMI test condition was compared to the status eHMI, which can be seen as the basis of all eHMI designs. For further comparisons, the status eHMI served as the baseline and was compared to the eHMI variants conveying additionally vehicle's perception and/or intent. Each model was tested in two analysis steps, using the software HLM Version 7.03: (1) First, an unconstrained (null) model was computed to test whether there is systematic variance in the outcome variable and an HLM is justified (Woltman, Feldstain, MacKay, & Rocchi, 2012). The intra-class correlation (ICC) represents the ratio (in %) of the total variance in the outcome variable that is attributable to the participants (level-2). As recommended by Lorah (2018), we used the ICC as the standardized effect size index for the level-2 factor participants. It can be interpreted in the same way as a correlation coefficient (Lorah, 2018), thus as $r = 0.1$ (small effect), $r = 0.3$ (medium effect), and $r = 0.5$ (large effect). If the variability in the outcome variable differs significantly from zero (determined by a Chi-square test), (2) secondly, the calculation of a random coefficients model is justified. We calculated Pearson's *r* as a standardized effect size measure for the level-1 factor eHMI, too, as recommended by Woltman et al. (2012).

Data analysis of the structured interview followed a qualitative content analysis based on Mayring (2014), which is characterized by a systematic procedure. Since the analysis is of explorative nature, inductive category formation was chosen as the analytic technique. The aim of inductive category formation is to summarize and derive categories based on the spoken material itself. We defined the level of abstraction as the concrete factors of a person, in a form that can also occur in other interviews; that is, idiosyncratic formulations were excluded. Answers to each light signal were analyzed separately by a step-by-step procedure of reduction and abstraction. Thus, a condensed category system is reported for each conceivable information display, that is, communicating the vehicle's status of driving in automated mode, its perception of the pedestrian near the vehicle, or its intent for the next maneuver. Each category system reflects benefits and limitations on three levels: First, the affective level contains the evoked emotions and subjective feelings stated by the subjects. Second, the cognitive level encompasses participants' general ideas, thoughts, and beliefs. Thirdly, the conative level addresses impacts on pedestrians' behavior. To this end, the interviews were recorded on tape and transcribed. The total scope of the logs was 15,891 words and was evaluated by two coders. The average inter-coder reliability was Cohen's $\kappa = 0.72$ – 0.82 , hence good to excellent (Cicchetti, 1994). All codings with disagreement were discussed and coders agreed upon one coding for the reported final category system.

3. Results

3.1. Trust, perceived safety, user experience, perceived intelligence, and transparency

Table 1 shows the results of the mixed design ANOVAs. All subjective measures revealed the same pattern: A main effect of eHMI was found, indicating that the displayed information by an eHMI impacts pedestrians' trust in SDVs, perceived

Table 1
Mixed design ANOVAs.

Measurement	Effect	df ₁	df ₂	F-value	p-value	η^2_c
Cognitive Trust	eHMI	2.66	151.32	32.06	$p < .001^{***}$	0.325
	traffic scenario	1	57	0.94	$p = .336$	
	eHMI $\times$ traffic scenario	2.66	151.32	0.85	$p = .458$	
Affective Trust	eHMI	2.99	170.55	20.30	$p < .001^{***}$	0.232
	traffic scenario	1	57	0.07	$p = .791$	
	eHMI $\times$ traffic scenario	2.99	170.55	0.62	$p = .601$	
Perceived Safety	eHMI	2.90	165.40	26.89	$p < .001^{***}$	0.282
	traffic scenario	1	57	0.11	$p = .738$	
	eHMI $\times$ traffic scenario	2.90	165.40	1.10	$p = .350$	
User Experience	eHMI	2.83	161.25	31.22	$p < .001^{***}$	0.323
	traffic scenario	1	57	0.38	$p = .542$	
	eHMI $\times$ traffic scenario	2.83	161.25	0.43	$p = .722$	
Perceived Intelligence	eHMI	2.60	147.95	32.55	$p < .001^{***}$	0.320
	traffic scenario	1	57	0.51	$p = .479$	
	eHMI $\times$ traffic scenario	2.60	147.95	0.53	$p = .640$	
Transparency	eHMI	3.58	204.13	53.07	$p < .001^{***}$	0.448
	traffic scenario	1	57	0.35	$p = .557$	
	eHMI $\times$ traffic scenario	3.58	204.13	0.08	$p = .982$	

* $p < .05$, ** $p < .01$, *** $p < .001$.

safety and user experience when interacting with an SDV, and perceived intelligence and transparency of SDVs, all $ps < 0.001$, $\eta^2_c = 0.232$ – 0.448 . There was no significant main effect of the traffic scenario, all $ps > 0.05$, indicating that evaluations are consistent among traffic scenarios, and there was no significant interaction between the eHMI and the traffic scenario, all $ps > 0.05$.

Fig. 4 presents pedestrians' subjective ratings of all test conditions. Since there were no significant differences found among traffic scenarios (intersection, parking lot), we decided to present the data aggregated over both testing groups. Based on Hinderks, Schrepp, and Thomaschewski (2019), the user experience scores can be interpreted as bad for no eHMI, above average for status and status + perception eHMI, and excellent for status + intent and status + perception + intent eHMI.

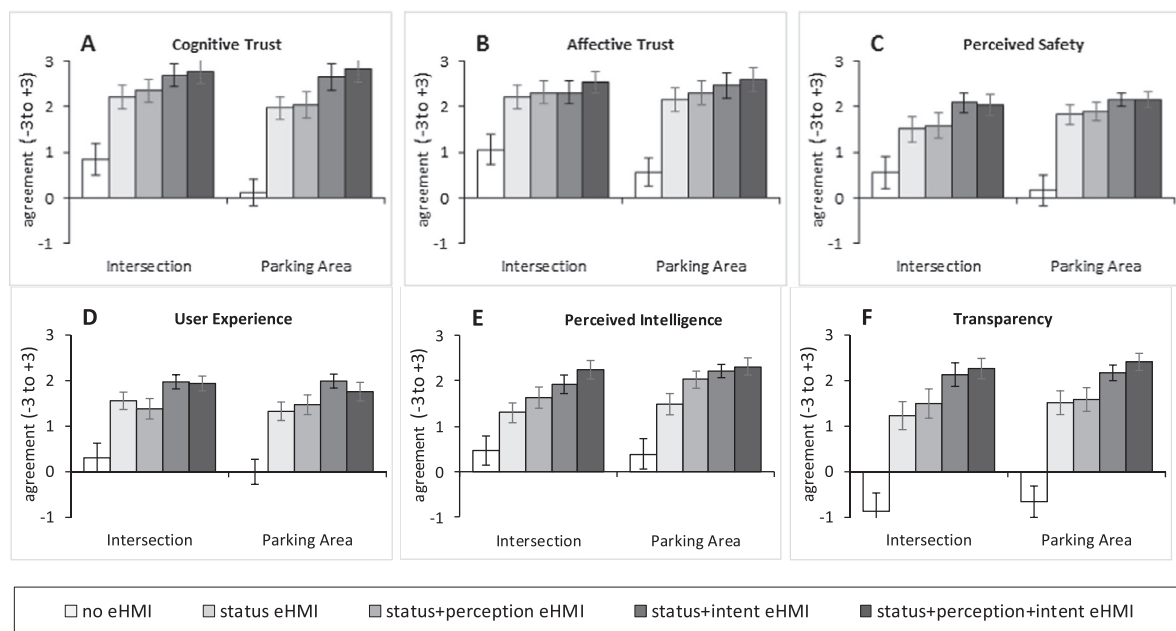


Fig. 4. Mean ratings of (A) cognitive trust, (B) affective trust, (C) perceived safety, (D) user experience, (E) perceived intelligence, and (F) transparency ratings. Error bars: ± 1 SE.

Post-hoc tests with Bonferroni corrections revealed that, on all measures, any eHMI displaying *some* information (status eHMI, status + perception eHMI, status + intent eHMI, status + perception + intent eHMI) is rated higher than no eHMI, all $ps < 0.001$, $r = 0.52$ – 0.83 .

When looking at the different eHMI information displays, there are no significant differences regarding affective trust and perceived safety, all $ps > 0.05$. However, on cognitive trust, the status + perception + intent eHMI is rated significantly higher than the status eHMI, $p < .05$, $r = 0.37$. On user experience, the status + intent eHMI is rated significantly higher than the status eHMI, $p < .01$, $r = 0.43$, and the status + perception eHMI, $p < .01$, $r = 0.44$. Regarding transparency, the status + perception + intent eHMI is rated significantly higher than the status eHMI, $p < .001$, $r = 0.58$, and the status + perception eHMI, $p < .01$, $r = 0.47$. Furthermore, the status + intent eHMI is perceived to be significantly more transparent than the status eHMI, $p < .05$, $r = 0.40$. With regard to perceived intelligence, the status + perception + intent eHMI is rated significantly higher than the status eHMI, $p < .001$, $r = 0.61$, and the status + perception eHMI, $p < .05$, $r = 0.40$. Participants perceive the status + intent eHMI to be significantly more intelligent than the status eHMI, $p < .001$, $r = 0.54$. All other comparisons are non-significant, all $ps > 0.05$.

Taken together, the status + perception eHMI is not rated higher than the mere status eHMI on any measure. The status + intent eHMI was rated higher than the mere status eHMI on user experience, perceived intelligence, and transparency, and higher than the status + perception eHMI on user experience. The status + perception + intent eHMI was rated higher than the status eHMI on cognitive trust, transparency, and perceived intelligence, and higher than the status + perception eHMI on transparency and perceived intelligence, but not rated higher than the status + intent eHMI on any measure.

3.2. Objective behavior

Fig. 5 shows the mean times and standard errors for the crossing/clearing onset and the crossing/clearing duration for each eHMI condition. Regarding the HLMs, all unconstrained (null) models were significant, all $ps < 0.01$, except for the clearing duration model status vs. status + perception, $p = .225$ (ICC: 9%). The results of the subsequent random coefficient models are displayed in Table 2. On all measures, the person variable was significant, indicating that pedestrians differ in their crossing and clearing strategies, all $ps < 0.001$, $r = 0.27$ – 0.75 .

Regarding crossing onset, all mean values are positive, indicating that participants tended to enter the street while the vehicle was still moving and did not wait until it came to a complete standstill. On a descriptive level, participants started crossing the street earliest with the status + intent eHMI before the vehicle came to a standstill and waited the longest to initiate crossing with no eHMI. However, the results of the HLMs indicate no significant difference regarding crossing onset, all $ps > 0.05$. On a descriptive level, crossing duration increases with the amount of information provided by the eHMI. Indeed, the HLMs show that pedestrians take significantly longer to cross the street with a status + perception + intent than with a status eHMI, $b = 0.11$, $p < .05$, $r = 0.31$.

On a descriptive level, clearing onset is earliest with a status + intent eHMI and latest for status + perception eHMI. Indeed, clearing onset is significantly later for the status + perception eHMI than the mere status eHMI, $b = 1.03$, $p = .05$, $r = 0.22$, hence people take longer to initiate leaving the vehicle's trajectory after the activation of a status + perception eHMI compared to a mere status eHMI. Descriptive values for clearing duration lay in a very narrow range, between $M = 1.15$ s for no eHMI and $M = 1.27$ s for status + intent eHMI. The HLMs show no significant differences between the eHMI test conditions regarding clearing duration.

3.3. Qualitative content analysis

When asked which information SDVs should display, 88.1% ($N = 52$) of the participants want to be informed if a vehicle is driving in automated driving mode, 55.9% ($N = 33$) of the participants prefer to receive information on the vehicle's

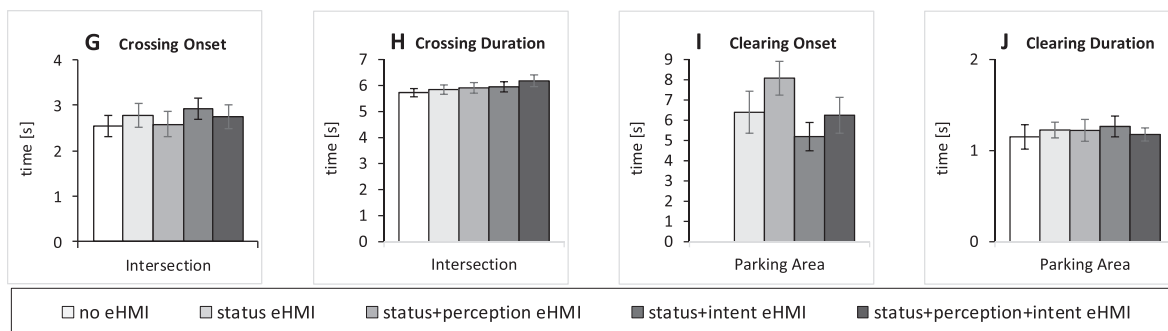


Fig. 5. Mean ratings of (G) crossing onset (higher values indicate higher traffic flow), (H) crossing duration, (I) clearing onset, and (J) clearing duration (lower values indicate higher traffic flow). Error bars: ± 1 SE. For clearing onset the no-eHMI test condition could not be computed due to technical difficulties.

Table 2
Hierarchical linear models.

Measurement	Test condition	Effect	b	SE	t-value	df	p-value	r
Crossing Onset	(1) vs. (2)	person	2.65	0.23	11.75	29	$p < .001$	0.72
		eHMI	0.23	0.18	1.25	29	$p = .221$	
	(2) vs. (3)	person	2.67	0.24	11.10	29	$p < .001$	0.60
		eHMI	−0.10	0.12	−0.79	29	$p = .437$	
	(2) vs. (4)	person	2.84	0.22	13.00	29	$p < .001$	0.53
		eHMI	0.15	0.24	0.61	29	$p = .547$	
	(2) vs. (5)	person	2.76	0.22	12.36	29	$p < .001$	0.49
		eHMI	−0.01	0.09	−0.06	29	$p = .951$	
	(1) vs. (2)	person	5.80	0.16	36.40	29	$p < .001$	0.75
		eHMI	0.09	0.12	0.73	29	$p = .473$	
Crossing Duration	(2) vs. (3)	person	5.88	0.18	33.24	29	$p < .001$	0.71
		eHMI	0.03	0.07	0.45	29	$p = .658$	
	(2) vs. (4)	person	5.90	0.18	33.52	29	$p < .001$	0.63
		eHMI	0.11	0.12	0.88	29	$p = .389$	
	(2) vs. (5)	person	6.01	0.19	32.38	29	$p < .001$	0.65
		eHMI	0.11	0.05	2.10	29	$p < .95^*$	
	(2) vs. (3)	person	7.55	0.89	8.44	26	$p < .001$	0.46
		eHMI	1.21	0.49	2.46	26	$p < .05^*$	
	(2) vs. (4)	person	5.89	0.73	8.02	27	$p < .001$	0.38
		eHMI	−1.46	1.01	−1.44	27	$p = .161$	
Clearing Onset	(2) vs. (5)	person	6.67	0.88	7.55	26	$p < .001$	0.60
		eHMI	0.09	0.31	0.29	26	$p = .774$	
	(1) vs. (2)	person	1.19	0.09	12.64	27	$p < .001$	0.44
		eHMI	0.10	0.12	0.86	27	$p = .397$	
	(2) vs. (4)	person	1.25	0.08	15.72	27	$p < .001$	0.27
		eHMI	0.00	0.13	−0.03	27	$p = .975$	
	(2) vs. (5)	person	1.20	0.06	18.83	26	$p < .001$	0.32
		eHMI	−0.02	0.03	−0.70	26	$p = .489$	

* $p < .05$, ** $p < .01$, *** $p < .001$. eHMI information: (1) no eHMI, (2) status, (3) status + perception, (4) status + intent, (5) status + perception + intent. Due to technical difficulties, the random coefficient model (1) vs. (2) for clearing onset could not be computed. The random coefficient model (2) vs. (3) for clearing duration was not computed, as the unconstrained null model was not significant.

perception, and 83.1% ($N = 49$) of the interviewees would like to receive information about the vehicle's intent. The two test groups "intersection" and "parking lot" did not differ significantly in their answers regarding information on an SDV's status, $\chi^2(1) = 0.13$, $p = .723$, perception, $\chi^2(1) = 0.01$, $p = .908$, or intent, $\chi^2(1) = 0.57$, $p = .451$, indicating that informational needs are consistent among traffic scenarios. Thus, we decided to present the data aggregated over both testing groups. Benefits and limitations expressed by participants that emerge from each conceivable information display are illustrated in Table 3. Frequency analysis of the derived category systems matches the picture drawn by the questionnaire data: For the status and intent displays, more codings in favor are stated (91.5% and 88.9% respectively) than against it (8.5% and 11.1% respectively). On the contrary, for the perception display, there are almost as many negative codings (43.9%) as positive ones (56.1%).

On the affective level, the most frequently stated category among all conceivable information displays is an improved *sense of safety*: "I feel much safer" (SP13). The status information is also positively associated with *no worries*, "that you are not frightened if you do not see a person in the car, because you can see that there is a system engaged" (SP52). Moreover, there are two opposing emotions in favor of a status indication, namely *trust in technology* and *lack of trust in technology*: Some people find a vehicle in automated driving mode more trustworthy than a vehicle steered by a human driver, whereas others want the information because they do not trust the ADS technology and would rather evade SDVs. On the other hand, participants raise concerns that informing about the vehicle's automated driving mode might lead to *overtrust* in the vehicle's capabilities among pedestrians, who might then be less attentive when encountering such a vehicle. For the perception and intent signal, people find that the information enhances their *trust in technology* and gives them a *pleasant feeling*. However, the perception signal also evokes negative feelings such as *feeling like a target*, *irritation*, and *feeling uneasy*: "With the eyes, I felt controlled, observed, and uneasy." (SP40) The category *trust in technology* is mentioned in favor of the intent information, too, since some participants find that the additional feedback enhances their feeling of trust in SDVs, whereas others argue that they do not need further signaling because they reflect *overtrust* in the ADS. "If I see that the automated driving mode is turned on, then I trust the technology." (SP13)

On the cognitive level, the most frequently stated category among all conceivable signals is *informedness* about the automated driving mode, vehicle's perception, and vehicle's intent, for example, "So that I and other road users are simply warned or informed that this is self-driving." (SP27) Regarding the driving mode indicator, many participants refer to the *changed role of driver*. One participant reasons this way: "Because I don't see a driver [...] And if, for example, someone would be sitting in the driver's seat, I would try to make contact with him, and if he doesn't react, maybe he reads a newspaper, I would be confused. That's why it helps me if the vehicle tells me: Listen, I am driving automated." (SP39) Some participants

Table 3

Reasons in favor of and against the conceivable eHMI information displays.

eHMI information	Level	Benefits			Limitations		
		Category	C	% of C	Category	C	% of C
Status	Affective	Total	130	91.5%	Total	12	8.5%
		Sense of safety	15	10.6%			
		No worry	9	6.3%			
		Lack of trust in technology	9	6.3%			
	Cognitive	Trust in technology	5	3.5%	Overtrust	2	1.4%
		Informedness	35	24.6%	Irrelevance	5	3.5%
		Changed driver's role	28	19.7%	Vehicle-centric cues	5	3.5%
		Market entry	13	9.2%			
	Conative	No technical defect	6	4.2%			
		Changed driving style	2	1.4%			
		Cautious behavior	8	5.6%			
Perception	Affective	Total	55	56.1%	Total	43	43.9%
		Sense of safety	12	12.2%	Feeling like a target	3	3.3%
		Trust in technology	5	5.4%	Irritation	2	2.2%
	Cognitive	Pleasant feeling	2	2.0%	Feeling uneasy	2	2.2%
		Informedness	22	23.9%	Irrelevance	20	21.7%
		Additional information	3	3.3%	Obvious ADS capability	7	7.6%
				Gimmick	4	4.3%	
	Conative			Information overload	3	3.3%	
		Replacement for communication with driver	7	7.6%			
		Interaction with intelligent system	2	2.2%			
		Trigger for own behavior	2	2.2%			
Intent	Affective	Total	121	88.9%	Total	17	11.1%
		Sense of safety	15	10.9%			
		Trust in technology	4	2.9%	Overtrust	2	1.4%
		Pleasant feeling	2	1.4%			
	Cognitive	Informedness	26	18.8%	Irrelevance	8	5.8%
		Additional information	15	10.9%	Vehicle-centric cues	5	3.6%
		Feeling detected	12	8.7%			
		Attention	9	6.5%			
	Conative	Similarity to rear brake lights	4	2.9%			
		Trigger for own behavior	16	11.6%			
		Replacement for communication with driver	7	5.1%			
	Time efficiency	7	5.1%				

C = no. of codings (reflects the number of participants in which a category had been coded), % = no. of category codings over all codings, calculated for each conceivable information display.

argue that the status indicator serves as a signal for *no technical defect*, as shown in the following exemplary statements: “It might happen that the vehicle drives without anyone on the driver's seat and then you are frightened, because you rather expect a technical defect and not that it is intended this way [...] That's why it makes sense to provide some kind of indication.” (SP04) “When the car moves, I have to know that it is moving in a controlled manner. Not that there is no handbrake and it just starts rolling. Because then I would try to stop the vehicle. The light makes me feel that the car is driving in a controlled manner.” (SP14) Apart from that, some participants find the status information particularly important for *market entry*, when people are not yet experienced with driverless vehicles.

While some participants appreciate the intent signal as *additional information* complementing the status information, only a few do so with the perception signal. Many participants feel that the perception signal is *irrelevant* while this is only true for a few participants regarding the status and intent signal. “I think you do not really need it,” (SP43) “I just think it's superfluous.” (SP03) Participants feel that the perception signal is an *obvious ADS capability*: “[...] Whether it perceives me or not – it was not so important for me. I'm assuming that such a car does that.” (SP42) Participants feel that the perception signal constitutes a *gimmick* and an *information overload*. “That's a nice gimmick, but I find it trivial.” (SP35) “I think it is too much.” (SP59) Participants already *feel detected* with the intent display; that is, they have the feeling that the car reacts to them. “The automated vehicle has recognized me and reacts to me: Hello. I saw you, I'm braking.” (SP24) “The car has recognized me and now initiates the necessary steps so I won't get hurt.” (SP59) Furthermore, participants like that the intent signal draws *attention*, and participants in the intersection scenario mention the benefit of the signal's *similarity to rear brake lights*. On the other hand, some participants argue that they do not need an eHMI showing status and intent, because *vehicle-centric cues* such as motion, light, or engine sounds of the vehicle suffice: “I see relatively quickly if someone is braking or not.” (SP35) “When the engine is on, I know that the vehicle wants to drive off.” (SP28)

On the conative level, some participants report that they would *behave cautiously* if the status information were indicated. “I would perhaps keep more distance, depending on how new the technology is.” (SP30) Both the perception and intent signal are perceived as a *replacement for communication with the driver*. “Just as the driver would look at me when I stand at the

crosswalk.” (SP06) “Normally, you can already tell by how someone is going to his car whether he wants to drive off right away [...]. Then when people have the steering wheel in their hands and look to the left and to the right to check if someone is approaching, maybe you can even see that they are engaging a gear. Or they are just leaning back, tapping on their smart-phone. Somehow you can tell from their behavior whether they want to drive off.” (SP53) Only few participants state further benefits of the perception signal, that is, *interacting with an intelligent system*. “This gives the impression that it is something intelligent and sophisticated, that moves, that is complex, [...] that registers me, that saw me.” (SP14) Participants report that the information on the vehicle’s intent is a *trigger for their own behavior*, and, consequently, associated with *time efficiency* as shown in the following statement for the intersection traffic scenario: “This way, I know that I can cross the street. Otherwise, I would wait for the vehicle to stop. It also saves time, [...] everything just goes a bit faster.” (SP40).

4. Discussion

The present study investigated whether pedestrians want to receive information from an SDV, and if so, which information they prefer to receive. This was achieved by exploring different eHMI variants: (1) no eHMI, (2) status, (3) status + perception, (4) status + intent, and (5) status + perception + intent. In a mixed design, participants acting as pedestrians experienced the five eHMI designs (within-subject factor) at either an intersection (when crossing the street in front of an approaching SDV) or a parking lot (when blocking an SDV that is planning to go) (between-subject factor).

Results showed that any eHMI leads to higher ratings of trust in SDVs, perceived safety and user experience when interacting with SDVs, as well as perceived intelligence and transparency of SDVs, compared to no eHMI, consistent among traffic scenarios. These beneficial effects are mainly driven by the status indication. In post-experimental interviews, participants reported that they want to receive a status indication most prominently due to the changed role of the driver. Adding a perception eHMI to the status indication does not lead to any further improvement. Communicating perception is perceived as obvious information (“gimmick”), even resulting in a later clearing onset and longer crossing duration for pedestrian behavior. In contrast, additional intent information further improves user experience, perceived intelligence, and transparency than a mere status indicator, but not trust or perceived safety. Pedestrians feel detected and more informed. Though some participants report that the intent signal triggers their behavior, objective behavioral measures showed no significant effects regarding traffic flow.

4.1. Informational need via an eHMI

The immediate finding from the analysis is that all eHMI variants support the interaction between pedestrians and SDVs in urban traffic. In the presence of any eHMI, pedestrians were more trusting and felt safer when encountering a SDV, which is in line with previous studies (e.g. Böckle et al., 2017; Chang et al., 2017; de Clercq et al., 2019; Habibovic et al., 2019). Moreover, pedestrians report an above-average (status, status + perception) or excellent (status + intent, status + perception + intent) user experience with an eHMI compared to a bad user experience if encountering an SDV without an eHMI. With any eHMI, pedestrians perceive SDVs to be more intelligent and transparent. Thus, the positive effect of providing information in human-machine interaction found in previous studies (e.g. Fischer et al., 2018) also holds for the interaction between pedestrians and SDVs. Subjective evaluations of the eHMI variants were consistent among the intersection and the parking lot traffic scenarios, allowing generalized conclusions.

The second immediate finding is that, with regard to the question about *which* information shall be provided by an eHMI, the status indicator is largely accountable for the positive eHMI effects. When interacting with an SDV, status information on the vehicle’s driving mode can be seen as the most basic information and was indeed recently recommended by the SAE (2019). If the automated driving mode is indicated, pedestrians can adapt their expectations of the driver’s role and the vehicle’s capabilities. In the post-experimental interview, nearly 90% of the participants reported that they want to be informed about the vehicle’s automated status. According to the participants, a main motivation for this informational need is the changed role of the driver in SDVs. Thus, the present study adds evidence to previous studies (e.g. Lagström & Lundgren, 2015; Lundgren et al., 2017; Rodríguez, 2017) showing that the absence of communication with the driver leads to an unsafe feeling among pedestrians, at least on a subjective level. An eHMI helps to meet pedestrians’ new communication needs that arise according to Lundgren et al. (2017) with an increasing level of vehicle automation. Some participants report that receiving this information is especially important during market entry, until the assumption of a driver steering the vehicle wanes and automated vehicles become more and more integrated in traffic.

The present study explored whether, besides a mere status eHMI, supplementary information on the vehicle’s perception and/or intent further improves the interaction between pedestrians and SDVs.

The findings indicate that additionally communicating the vehicle’s perception is neutral at best. With regard to the subjective measures, status + perception is never rated significantly better than a mere status indicator. Similarly, status + perception + intent does not reflect higher ratings than the status + intent eHMI for any measure. Communicating perception polarized the participants, with almost as many participants reporting that they do not want to be informed about the vehicle’s perception as participants in favor of it. This manifests itself in the contrasting impressions, where many of the arguments in favor of perception find a negative counterpart: *pleasant feeling* vs. *feeling uneasy*, *informedness about vehicle’s perception* vs. *irrelevance*, *additional information* vs. *information overload*, and *interaction with intelligent car* vs. *gimmick*. While

feeling uneasy is possibly a conceptual concern of the specific tested design of moving stylized eyes, resulting from the eeriness people experience with anthropomorphic robots as suggested by the uncanny valley hypothesis (Mori, MacDorman, & Kageki, 2012), most negative concerns are raised on a more abstract level. For example, given its sensors, participants regard it as obvious that an SDV perceives them, making the display an irrelevant gimmick. The reported information overload is also reflected in the significant longer times to clear the vehicle's trajectory on the parking lot, and to cross the intersection, compared to when only the status information is indicated. These results suggest that providing pedestrians with information on the vehicle's perception catches pedestrians' attention and distracts them in a way that traffic flow is negatively impacted. For an optimal eHMI design, it is important to find a balance between informing yet not overwhelming other road users, as emphasized by the ISO (2018).

In contrast to that, the present study found evidence that providing information on the vehicle's intent potentially improves the interaction between pedestrians and SDVs. Participants rated user experience, perceived intelligence, and transparency of the SDV higher when encountering the vehicle with the status + intent eHMI compared to the mere status information. However, trust and perceived safety are not further improved by an additional intent signal. About four-fifths of the participants reported that they want to be informed about the vehicle's intent. Participants reported that the intent cue served as a trigger for their behavior and even increases their time efficiency. However, objective measures were not able to show any significant effects. Apart from that, participants report that the intent signal already implies that the vehicle has detected a pedestrian. Participants assume that the SDV displays its intent as a reaction to a pedestrian being detected.

Thus, the third immediate finding is that informing pedestrians about the vehicle's intent is more appropriate additional information than informing pedestrians about the vehicle's perception.

Furthermore, the present study found evidence that an eHMI is not the main factor influencing pedestrians' behavior when interacting with an SDV. Findings of the HLM reveal that differences in crossing and clearing behavior are largely due to the person level. This supports the assumption of Clamann et al. (2017), who state that pedestrians use individual crossing strategies.

4.2. Limitations and future research

A major limitation of the current study is its predictable and artificial testing setup, which is not representative for everyday traffic scenarios. In particular, pedestrians stood at a fixed point and the study always guaranteed the participant's safety, meaning that the vehicle always stopped when the pedestrian was in or near the vehicle's trajectory. However, a naturalistic study on a testing site cannot incorporate any non-yielding vehicles or mixed traffic due to ethical standards. At the same time, this procedure allowed for controlled and replicable test conditions. Future simulation studies (such as video studies or VR studies) could integrate non-yielding vehicles, mixed traffic, and/or a malfunctioning eHMI test conditions. A further limitation goes back to the fact that the meaning of the eHMI light concepts was explained to the participants before their vehicle encounters. This approach might have introduced some demand characteristics, potentially biasing and priming participants into paying attention to the light elements. Participants could expect information to be displayed, which might have diverted them from other critical points. Nevertheless, we did so because previous studies showed that participants have trouble guessing the meaning of light signals without explanation (Hensch et al., 2020), but are successful in their interpretation after explanation (e.g. Habibovic et al., 2018). Thus, it can be assumed that the meaning of a light-based eHMI is not self-explanatory but has to be learned as a convention, analogous to other vehicle lights such as turn indicators and brake lights. The present study focused on pedestrians' informational needs when encountering SDVs, but not on the intuitiveness of eHMI designs. Future ideation workshops should target how these informational needs can be displayed in an intuitive design, preferably with participants from different cultural backgrounds. Apart from that, it should be noted that the chosen eHMI design might had a confounding influence on pedestrians' preferences for a specific message. In the current study, we tried to mitigate these effects by limiting the eHMI design space to one modality and type of signal for the different displayed information.

Moreover, another limitation is that study participants encountered every eHMI only once. We cannot draw conclusions about long-term effects of eHMIs conveying specific information and how pedestrians' expectations to be informed by SDVs will evolve. Future studies should investigate whether an eHMI displaying status information is sufficient regarding pedestrians' informational needs or whether the intent information is vital for a good eHMI. A long-term study can show to what extent the observed effects persist with increasing experience beyond the first encounter. In addition, it is easy to imagine that the self-reported triggering effect of an intent eHMI requires habituation, thus might exhibit its positive effect on traffic flow after multiple trials.

5. Conclusions

In summary, this study provides evidence that pedestrians are more trusting, feel safer, rate user experience higher, and perceive SDVs as more intelligent and transparent if equipped with an eHMI that provides at least information on the vehicle's ADS status. Our work shows that informing pedestrians additionally about the vehicle's perception does not provide any added value, is perceived as an obvious gimmick, and even has a negative impact on traffic flow. In contrast, additional information on the vehicle's intent leads to subsequent improvements in user experience, perceived intelligence, and

transparency compared to when only status information is conveyed. The observed informational needs were stable among different low-speed urban traffic scenarios.

Thus, it can be concluded that pedestrians' informational needs lie in receiving information about the vehicle's ADS status and, possibly, vehicle's intent, but not about the vehicle's perception. Future research should address long-term evaluations of the status and intent eHMI to decide whether a status eHMI is sufficient for pedestrian-SDV interaction or whether a subsequent intent eHMI is vital.

Furthermore, the present study provides evidence that pedestrians show highly individual crossing and clearing strategies.

CRediT authorship contribution statement

Stefanie M. Faas: Conceptualization, Methodology, Software, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review & editing, Visualization, Supervision, Project administration. **Lesley-Ann Mathis:** Methodology, Formal analysis, Investigation, Writing - original draft, Writing - review & editing, Visualization. **Martin Baumann:** Conceptualization.

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